

Project Executable Files

Date	15 July 2026
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

This section documents all executable and deployable files for Rising Waters, so an evaluator can run or deploy the solution using the provided files and instructions.

Step 1: Submission Checklist

S. No	Item to Submit	Submitted
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	N/A — no database used; flat-file model artifacts only
5	Environment configuration file (.env.example or similar)	N/A — no API keys or secrets required
6	Deployed application URL (if hosted)	Yes — https://rising-waters-12fj.onrender.com/
7	APK / Executable binary (if applicable)	N/A — web app only, no mobile/desktop binary
8	Dockerfile / Containerization config (if applicable)	N/A — deployed via Render's native Python buildpack
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Pending — to be recorded before final submission

Step 2: File / Folder Structure

Overview of the Rising Waters file and folder structure:

```
Rising-Waters/  
  core/ - Validation and prediction logic  
  dataset/ - Processed training data (flood_dataset.csv)  
  models/ - Saved model + scaler (floods.save, transform.save)  
  notebooks/ - Exploratory analysis / training notebook  
  raw_data/ - Source rainfall dataset  
  static/ - CSS and JavaScript assets  
  templates/ - home.html, index.html, chance.html, nochance.html  
  tests/ - Automated test suite  
  app.py - Flask application entry point  
  train.py - Model training pipeline  
  requirements.txt - Python dependencies  
  Readme.md - Setup and run instructions
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://rising-waters-12fj.onrender.com/
Login Credentials (Demo)	Not applicable — public, anonymous access, no login required
Platform / Hosting Provider	Render
Repository Link	https://github.com/NandaGunasri/Rising-Waters
Demo Video Link	[To be added before final submission]

Step 4: Run Instructions

Step-by-step instructions for an evaluator to run the project locally:

Pre-requisites: Python 3.11+, pip

Step 1: git clone <https://github.com/NandaGunasri/Rising-Waters>

Step 2: pip install -r requirements.txt

Step 3: python train.py (builds the model — first time only)

Step 4: python app.py

Step 5: Open a browser at <http://127.0.0.1:5000/>

Step 5: Known Issues / Limitations

S. No	Known Issue / Limitation	Workaround / Status
1	Free-tier Render hosting causes a cold-start delay after inactivity	First request after an idle period may take 10–20 sec to respond; subsequent requests are fast
2	Cloud Cover is a derived proxy when training on the public rainfall dataset (no real cloud-cover column)	Documented in the Readme; acceptable for demo purposes

S. No	Known Issue / Limitation	Workaround / Status
3	No live weather API integration yet	Planned for Phase 2 roadmap (NASA POWER / Copernicus)